SPECTRAL INEQUALITIES FROM BLACK-BOX RESULTS II

FELIPE LABRA

Abstract.

Contents

1. Review from past session 1
1.1. Black-Box I: Carleman Estimates 1
2. Black-Box II: Propagation of smallness 3
2.1. Control for heat equations on "very small sets" 8
3. Black-Box III: Unbounded domain case 14
3.1. Setting 14
3.2. Principal results 14
3.3. Proof of Theorem 1 16
4. 6/8: Double manifold and further results 19
4.1. Black-Box IV: The double manifold 19
4.2. Simultaneous control for the Heat equation with Dirichlet and Neumann boundary conditions 21
4.3. Control and the double manifold 22
References 24

1. Review from past session

Suppose that $\Omega \subset \mathbb{R}^d$ is a smooth bounded open subset. By the spectral theorem, there exists an spectral decomposition for the Dirichlet Laplacian $-\Delta$ given by $(e_k, \lambda_k^2)_k \in L^2(\Omega) \times \mathbb{R}^+$ formed by an orthonormal basis of $L^2(\Omega)$ and $\lambda_k^2 \to +\infty$.

The previous trick was known as the *elliptic lift*. For $\phi = \sum_{\lambda_k \leq \Lambda} u_k e_k$, we write

$$u(s,x) = \sum_{\lambda_k \leq \Lambda} \frac{\sinh(s\lambda_k)}{\lambda_k} u_k e_k(x),$$

which satisfies

$$\begin{cases} (-\partial_s^2 - \Delta)u = 0, & \mathbb{R}_s \times \Omega \\ \partial_s u = \phi, & \Omega. \end{cases}$$

This shows that a spectral inequality

$$\|\phi\|_{L^2(\Omega)} \leq Ce^{c\Lambda} \|\phi\|_{L^2(\omega)},$$

is an unique continuation statement for u from $\{0\} \times \omega$.

1.1. Black-Box I: Carleman Estimates.

Theorem 1.1 ([JL99], Theorem 5.1)**.** *Let $\omega \subset \Omega$ ve a non-empty open set. There exists $K > 0$ such that for all $\Lambda > 0$ we have*

$$\|\phi\|_{L^2(\Omega)} \leq e^{K\Lambda} \|\phi\|_{L^2(\omega)}, \qquad \phi \in \operatorname{span}\{\lambda_k \mid \ \lambda_k \leq \Lambda\}.$$

Main idea of the proof: Use properly the elliptic lift. Let $S_0 > 0$ and $\beta \in (0, S_0/2)$. Let $Z := (0, S_0) \times \Omega$ amd set $Y = (\beta, S_0 - \beta) \times \Omega$. We set $z = (s, x)$ with $s \in (0, S_0)$ and $x \in \Omega$. We define the augmented elliptic operator

$$Q := D_s^2 + P_0 + R, \quad \text{in } Z,$$

where R is a first order differential operator in Z with bounded coefficients.

Theorem 1.2 ([RLR22], Theorem 7.12)**.** *Let $\omega \subset \Omega$ be an open subset. There exists $C > 0$ and $\delta \in (0, 1)$ such that $u \in H^2(Z)$ that satisfies $u\,|_{(0,S_0)\times\partial\Omega} = 0$ for $s \in (0, S_0)$ we have*

$$\|u\|_{H^1(Y)} \leq C\|u\|^{1-\delta}_{H^1(Z)} \left(\|Qu\|_{L^2(Z)} + |u\,|_{s=0}|_{H^1(\omega)} + |\partial_s u\,|_{s=0}|_{L^2(\omega)}\right)^{\delta}$$

2. Black-Box II: Propagation of smallness

The main idea is to consider a weaker assumption over $\omega \subset \mathbb{R}^d$, passing from an open set to just a set with positive Lebesgue measure.

The idea is to follow works by Burq-Moyano [BM23; BM25], which extends the spectral projector estimes from Jerison-Lebeau [JL99]. Part of the proof is inspired in results from propagation of smallness from wild sets due yo Logunov-Malinnikova [LM18].

Let $(\mathcal{M}, g)$ be a compact Riemannian manifold with boundary and let Δ be the Laplace operator on M,

$$\Delta = \frac{1}{\kappa(x)} \operatorname{div}(g^{-1}(x)\kappa(x)\Delta_x) = \operatorname{div}_\kappa \nabla_g,$$

where κ is a positive Lipschitz density. For instance, we recover the usual Laplace-Beltrami operator when $k(x) = |g|^{1/2}$. Denote by (e_k) the family of eigenfunctions of $-\Delta$ with eigenvalues $\lambda_j^2 \to +\infty$. Then

$$\begin{cases} -\Delta e_k = \lambda_k^2 e_k, & M, \\ \alpha e_k + \beta \partial_\nu e_k = 0, & \partial M, \end{cases}$$

where $\alpha, \beta \in \{0, 1\}$ and $\alpha + \beta = 1$. In what follows, we consier an arbitrary finite linear combination of the form

$$\phi = \sum_{\lambda_k \leq \Lambda} u_k e_k(x).$$

The manifold M is assumed $W^{2,\infty}$ with dimension d, or without boundary. This is needed for the $C^1 \cap W^{2,\infty}$ regularity for local charts.

Recall the $n-$Hausdorff measure of a set $E \subset \mathbb{R}^d$,

$$\mathcal{C}_{\mathcal{H}}^n(E) = \inf \left\{ \sum_j r_j^n \mid \ E \subset \cup_j B(x_j, r_j) \right\},$$

and the Hausdorff dimension of E defined as

$$\dim_{\mathcal{H}}(E) = \inf\{n \mid \ \mathcal{C}_{\mathcal{H}}^d(E) = 0\}.$$

Denote by $|E|$ the Lebesgue measure of E. The $d-$Hausdorff measure is equivalent to the $d-$Lebesgue measure: there exists $C_d, c_d > 0$ such that for every Borel set

$$c_d|A| \leq \mathcal{C}_{\mathcal{H}}^d(A) \leq C_d|A|,$$

and

$$\mathcal{C}_{\mathcal{H}}^d(E) > 0 \implies \forall d' < d, \ \mathcal{C}\mathcal{C}_{\mathcal{H}}^{d'}(E) \geq \inf\{1, \mathcal{C}_{\mathcal{H}}^d(E)\}.$$

Remark 2.1. *The value of the Hausdorff measure is not invariant by diffeomorphisms, but it is by Lipschitz diffeomorphisms.*

The following result generalizes Jerison-Lebeau's work [JL99].

{Thm2.1}

Theorem 2.1 ([BM23], Theorem 1)**.** *There exists $\delta \in (0, 1)$, depending only on the dimension of the manifold, such that for any $m > 0$, there exists $C, D > 0$ such that for anu $E_1 \subset M$ with $|E_1| \geq m$ and any $E_2 \subset M$ sastisfying*

$$\mathcal{C}_{\mathcal{H}}^{d-\delta}(E_2) > m, \tag{2.1}$$

{{Eq2.1}}

and any $\Lambda > 0$, we have

$$\phi = \sum_{\lambda_k \leq \Lambda} u_k e_k(x) \implies \|\phi\|_{L^\infty(M)} \leq C e^{D\Lambda} \|\phi \mathbf{1}_{E_1}\|_{L^1(M)}, \tag{2.2}$$

{{Eq2.2}}

$$\phi = \sum_{\lambda_k \leq \Lambda} u_k e_k(x) \implies \|\phi\|_{L^\infty(M)} \leq C e^{D\Lambda} \sup_{x \in E_2} |\phi \mathbf{1}_{E_1}|, \tag{2.3}$$ {{Eq2.3}}

Remark 2.2. *The assumption (2.1) is not invariant under change of variables. It has to be understood in a fixed local chart (and the same applied for the theorem). One may use a condition* $\dim_{\mathcal{H}}(E) > d - \delta'$ *for* $0 < \delta' < \delta$*, which implies (2.1) and is invariant by Lipschitz diffeomorphisms.*

Remark 2.3. *Since the only requirement on* E_2 *is* $|E_2| > 0$*, the result applies even when* E_2 *is concentrated arbitrarily close to* ∂M*.*

These spectral inequalities naturally lead to observability estimates, hence controllability results. Note from (2.2) that if $F \subset (0,T) \times M$ has positive Lebesgue measure, there exists $C > 0$ such that for any $u_0 \in L^2(\mathcal{M})$, then

$$\|e^{T\Delta} u_0\|_{L^2(M)} \leq C \int_F |u(t,x)| \kappa(x)\, dxdt,$$

where $u = e^{t\Delta} u_0$ is a solution to the heat equation with Dirichlet or Neumann boundary condition. In the same spirit, from (2.5), there exists $\delta \in (0,1)$ depending on the dimension of M, such that for any $E \subset M$ with positive $(d-\delta)$-Hausdorff measure, and any $J \subset (0,T)$ of positive Lebesgue measure, there exists $C > 0$ such that for any $u_0 \in L^2(M)$, then

$$\|e^{T\Delta} u_0\|_{L^2(M)} \leq C \int_J \sup_{x \in E} |u(t,x)| dt.$$

This is summarized in the next result.

Theorem 2.2 ([BM23], Theorem 2. Null controllability from sets of positive measure)**.** *Let* $F \subset (0,T) \times M$ *of positive Lebesgue measure. Then there exists* $C > 0$ *such that for any* $u_0 \in L^2(M)$ *the solution* $u = e^{t\Delta} u_0$ *to the heat equation*

$$\begin{cases} (\partial_t - \Delta) u = 0, & (0,T) \times M, \\ u = 0 \text{ or } \partial_\nu u = 0, & (0,T) \times \partial M, \\ u(t,0) = u_0, & M. \end{cases}$$

satisfies

$$\|e^{T\Delta} u_0\|_{L^2(M)} \leq C \int_F |u(t,x)| \kappa(x)\, dxdt.$$

As a consequence, for all $u_0, v_0 \in L^2(M)$*, there exists* $f \in L^\infty(F)$ *such that the solution to*

$$\begin{cases} (\partial_t - \Delta) u = f \mathbf{1}_F(t,x), & (0,T) \times M, \\ u = 0 \text{ or } \partial_\nu u = 0, & (0,T) \times \partial M, \\ u(t,0) = u_0, & M, \end{cases}$$

satisfies $u \,|_{t=T} = e^{T\Delta} v_0$*.*

Theorem 2.3 ([BM23], Theorem 3. (Observability and exact controllability from measure zero sets))**.** *There exists* $\delta \in (0,1)$ *depending on the dimension of* M *such that for any* $E \subset M$ *with positive* $(d-\delta)$*-Hausdorff content and any* $J \subset (0,T)$ *of positive Lebesgue measure, there exists* $C > 0$ *such that for any* $u_0 \in L^2(M)$*, the solution* $u = e^{t\Delta} u_0$ *to*

$$\begin{cases} (\partial_t - \Delta) u = 0, & (0,T) \times M, \\ u = 0 \text{ or } \partial_\nu u = 0, & (0,T) \times \partial M, \end{cases}$$

satisfies

$$\|e^{T\Delta} u_0\|_{L^2(M)} \leq C \int_J \sup_{x \in E} |u(t,x)| dt.$$

As a consequence, assuming additionally that E is closed in M, for all $u_0, v_0 \in L^2(M)$ there exists a Borel measure μ supported on $(0,T)\times E$ such that the solution to

$$\begin{cases} (\partial_t - \Delta)u = \mu(t,x)\mathbf{1}_{J\times E}, & (0,T)\times M, \\ u = 0 \text{ or } \partial_\nu u = 0, & (0,T)\times \partial M, \\ u(0,x) = u_0, & M, \end{cases}$$

satisfies $u\,|_{t\geq T} = e^{t\Delta}v_0$.

The main tool to prove Theorem 2.1 is a very deep result on propagation of smallness for elliptic equations due to Logunov-Malinnikova [LM18]. Let u ve a solution of an elliptic equation in divergence form in a bounded domain $\Omega \subset \mathbb{R}^n$,

$$\operatorname{div}(A\nabla u) = 0,$$

where $A(x) = [a_{ij}(x)]_{1\leq i,j\leq n}$ is a symmetric uniformly elliptic matrix with Lipschitz entries,

$$\Lambda_1^{-1}\|\zeta\|^2 \leq \langle A\zeta,\zeta\rangle \leq \Lambda_1\|\zeta\|^2, \quad |a_{ij}(x) - a_{ij}(y)| \leq \Lambda_2|x-y|,$$

where $\Lambda_1 \geq 1$ and $\Lambda_2 \geq 0$.

Theorem 2.4 ([LM18], Theorem 5.1)**.** *There exists a constant $c \in (0,1)$ that depends only on the dimension n such that the following holds. Let $m, \delta, \rho > 0$ and suppose that $E, \mathcal{K} \subset \Omega$ satisfies*

$$\mathcal{C}_{\mathcal{H}}^{n-1-c+\delta}(E) > m, \quad \operatorname{dist}(E,\partial\Omega) > \rho, \quad \operatorname{dist}(\mathcal{K},\partial\Omega) > \rho.$$

Then there exist $C, \gamma > 0$ depending on $m, \delta, \rho, \Lambda_1, \Lambda_2, \Omega$ such that

$$\sup_{\mathcal{K}} |\nabla u| \leq C\left(\sup_E |\nabla u|\right)^{\gamma}\left(\sup_\Omega |\nabla u|\right)^{1-\gamma}.$$

Remark 2.4. *The previous theorem is a sharper result of a classic inequality known as* Three spheres theorem*: For solutions u to $Lu = 0$ in $\mathbb{R}^n$ where L is an uniformly elliptic operator, we have*

$$\sup_B |u| \leq C\left(\sup_{\frac{1}{2}B} |u|\right)^{\gamma}\left(\sup_{2B} |u|\right)^{1-\gamma},$$

where $B = \{x \in \mathbb{R}^n : |x| \leq 1\}$, and $C > 0, \gamma \in (0,1)$ depends only on L and not on u. In other words, we're trying to get the same estimation but on a lower level of regularity for the subsets $\frac{1}{2}B \subset B \subset 2B$.

To prove the others theorems we need to extend the results to the case of manifolds with boundary. This is done by using the double manifold, which is topologically two copies of the original manifold withe Dirichlet and Neumann conditions respectively, glued along the boundary. Hence these two copies embed isometrically in the double manifold. When the manifold is smooth, this is done by reflection across the boundary using geodesic coordinates. With lower regularity this is not possible, so a new coordinate system is developed. See [BM23, Section 5].

2.0.1. ***Very small sets implies non-zero measure sets.*** We want to prove that (2.5) implies (2.2). Set

$$\phi = \sum_{\lambda_k \leq \Lambda} u_k e_k(x), \quad \|\phi\|_{L^2(M)} = 1.$$

Let $F \subset M$ with $|F| > m/2$. Then from the equivalence

$$\mathcal{C}_{\mathcal{H}}^d(F) > c_d m/2, \quad \text{so} \quad \mathcal{C}_{\mathcal{H}}^{d-\delta}(F) \geq \min\{1, c_d m/2\}.$$

Now, from the spectral estimate [JL99, Theorem 14.6]

$$\|\phi\|_{L^\infty(M)} \leq Ce^{d\Lambda}\sup_{x\in F}|\phi(x)|, \tag{2.4}$$ {{Eq2.4}}

with uniform constants as long as $|F| \geq m/2$. Let

$$F = \left\{ x \in E_1 \mid |\phi(x)| \leq \frac{1}{2C} e^{-D\Lambda} \|\phi\|_{L^\infty(M)} \right\}.$$

If $|F| \geq m/2$, we have

$$\|\phi\|_{L^\infty(M)} \leq C e^{D\Lambda} \sup_{x \in F} \leq \frac{1}{2} \|\phi\|_{L^\infty(M)},$$

which shows that F cannot satisfy (2.4) as $\phi \neq 0$ since $\|\phi\|_{L^2(M)} = 1$. Hence $|F| < m/2$ and thus

$$\int_{E_1} |\phi(x)| dx \geq \int_{E_1 \setminus F} |\phi(x)| dx \geq \frac{|E_1|}{4C} e^{-D\Lambda} \|\phi\|_{L^\infty(\mathcal{M})},$$

which gives the desired estimate.

2.0.2. *Proof of the estimate [BM23], Theorem 1.* First we will check the boundaryless case. Let $m > 0$ and let $E_2 \subset M$ a given set with $\mathcal{C}_{\mathcal{H}}^{d-\delta}(E_2) > m$. First we localize on a local chart. Extract a finite covering

$$M \subset \cup_{j=1}^N U_j,$$

and $W^{2,\infty}$ diffeomorphisms $\psi_j : V_j \to \mathbb{R}^d$ with V_j being a neighbourhood of U_j in M. From [JL99, Theorem 14.6], there exists $C, D > 0$ depending only on M such that for any $j \in \{1, \dots, N\}$,

$$\|\phi\|_{L^2(M)} \leq C e^{d\Lambda} \|\phi\|_{L^2(U_j)}.$$

Let j_0 be such that

$$\mathcal{C}_{\mathcal{H}}^{d-\delta}(E_2 \cap U_{j_0}) > \frac{m}{N}.$$

We now work on the local chart U_{j_0} and define

$$V := \psi_{j_0}(V_{j_0}), \quad U := \psi_{j_0}(U_{j_0}), \quad F := \psi_{j_0}(E_2 \cap U_{j_0}).$$

Observe that ψ_{j_0} is a diffeomorphism of class $W^{2,\infty}$ by hypothesis, we must have

{{Eq2.3}}
$$\mathcal{C}_{\mathcal{H}}^{d-\delta}(F) > C(\psi_0) \frac{m}{N}. \tag{2.5}$$

In the chart $\psi_{j_0} : V_{j_0} \to V \subset \mathbb{R}^d$, the pushforward $(\psi_{j_0})_*$ of a function h on V is

$$((\psi_{j_0})_* h)(y) = h\left(\psi_{j_0}^{-1}(y)\right), \quad y \in V.$$

Now denote by $f_k = (\psi_{j_0})_* e_k$ and $\varphi = (\psi_{j_0})_* \phi$ the images of e_k and ϕ by the pushforward of ψ_{j_0} , which now are defined on V. This means that the equation $(-\Delta) e_k = \lambda_k^2 e_k$ on V_{j_0} transforms into an equation of the same form, with metric g and density κ as functions $W^{1,\infty}$ on $V \subset \mathbb{R}^d$,

$$\frac{1}{\kappa(x)} \operatorname{div}_x \left(\kappa(x) g^{-1}(x) \nabla_x f_k \right) = -\lambda_k^2 f_k, \quad \text{on } V.$$

Set

$$u(t,x) = \sum_{\lambda_k \leq \Lambda} \frac{\sinh(\lambda_k t)}{\lambda_k} u_k f_k(x),$$
$$\varphi(x) := \sum_{\lambda_k \leq \Lambda} u_k f_k(x) = \partial_t u \mid_{t=0}.$$

We have made the convention that $\sinh(\lambda t)/t = t$ for $\lambda = 0$. Then we have that

$$\left(\operatorname{div}(g^{-1} \kappa \nabla_x) + \partial_t \kappa \partial_t \right) u = 0.$$

For $T_2 > T_1 > 0$, let us define

$$E = \{0\} \times F, \quad K := [-T_1, T_1] \times \overline{U}, \quad \Omega = (-T_2, T_2) \times V,$$

which by construction satisfy the inclusions $E \subset K \subset \Omega$. Using (2.5) with $d = n-1$ and $\mathcal{C}_{\mathcal{H}}^{d-\delta}(E) = \mathcal{C}_{\mathcal{H}}^{d-\delta}(F)$,

$$\mathcal{C}_{\mathcal{H}}^{n-1-\delta}(E) > m', \quad n = d+1, \ m' = C(\psi_0)\frac{m}{N}.$$

For $\delta > 0$ small, we have from [LM18, Theorem 5.1],

$$\sup_K |\nabla_{t,x} u| \leq C \left(\sup_E |\nabla_{t,x} u|\right)^{\alpha} \left(\sup_\Omega |\nabla_{t,x} u|\right)^{1-\alpha}. \tag{2.6}$$ {{Eq2.5}}

We will use the following lemma.

Lemma 2.1. *[BM23] There exists $\sigma > 0$ such that*

$$\|\nabla_x u\|_{L^\infty} + \|u\|_{L^\infty} \leq C\|u\|_{\mathcal{H}^\sigma},$$

with $\mathcal{H}^\sigma := D((-\Delta)^{\sigma/2})$ endowed with $\|\phi\|_{\mathcal{H}^\sigma}^2 = \sum_k (1+\lambda_k)^{2\sigma}|u_k|^2$.

Remark 2.5. *$\mathcal{H}^\sigma$ does not necessarily coincides with the usual Sobolev space H^σ. When M is a compact manifold without boundary amd smooth metric both spaces coincide, so the lemma is just a Sobolev injection. In this level or regularity this is not the case, both spaces coincide when $0 \leq \sigma \leq 2$.*

First we have

$$\partial_t u(t,x) = \sum_{\lambda_k \leq \Lambda} \cosh(\lambda_k t) u_k f_k(x),$$
$$\nabla_x u(t,x) = \sum_{\lambda_k \leq \Lambda} \sinh(\lambda_k t) u_k \nabla f_k(x).$$

Then for $|t| \leq T_2$ and $\lambda_k \leq \Lambda$ we have $|\cosh(\lambda_k t)| \leq e^{T_2\Lambda}$ and $|\sinh(\lambda_k t)/\lambda_k| \leq T_2 e^{T_2\Lambda}$. Hence

$$\|\partial_t u(t,\cdot)\|_{\mathcal{H}^\sigma} = \sum_{\lambda_k \leq \Lambda} |u_k|^2 |\cosh(\lambda_k t)|^2 (1+\lambda_k)^{2\sigma} \leq (1+\Lambda) e^{2T_2\Lambda} \|\phi\|_{L^2(M)}^2,$$
$$\|u(t,\cdot)\|_{\mathcal{H}^\sigma}^2 \leq T_2^2 (1+\Lambda)^{2\sigma} e^{2T_2\Lambda} \|\phi\|_{L^2(M)}^2.$$

Recalling that $\Omega = (-T_2, T_2) \times V$ and using the previous Sobolev embedding,

$$\begin{aligned}\sup_{x\in\Omega} |\nabla_{t,x} u(t,\cdot)| &\leq C \sup_{t\in(-T_2,T_2)} (\|\partial_t u(t,\cdot)\|_{L^\infty} + \|\nabla_x u(t,\cdot)\|_{L^\infty}) \\ &\leq C(1+\Lambda)^\sigma e^{T_2\Lambda} \|\phi\|_{L^2(M)} \\ &\leq C' e^{(T_2+1)\Lambda} \|\phi\|_{L^2(M)}.\end{aligned}$$

By definition and recalling that $E = \{0\} \times F$,

$$\nabla_x u \mid_E = 0, \quad u \mid_E = 0, \quad \partial_t u \mid_E = \varphi \mathbf{1}_F.$$

Using the spectral estimate followed by the propagation of smallness result, both from [LM18],

$$\begin{aligned}\|\phi\|_{L^2(M)} &\leq C e^{D\Lambda} \|\phi\|_{L^2(U_j)} \\ &\leq C' e^{D\Lambda} \sup_{U_j} |\phi| \\ &\leq C' e^{D\Lambda} \sup_K |\nabla_{t,x} u| \\ &\leq C'' e^{D\Lambda} \left(\sup_E |\nabla_{t,x} u|\right)^{1-\alpha} \left(\sup_\Omega |\nabla_{t,x} u|\right)^{\alpha} \\ &\leq C'' e^{D\Lambda} \left(\sup_F |\phi|\right)^{\alpha} \|\phi\|_{L^2}^{1-\alpha},\end{aligned}$$

it follows that

$$\|\phi\|^{\alpha}_{L^2(M)} \leq C'' e^{D\Lambda} \left(\sup_F |\phi| \right)^{\alpha},$$

which after rising to the power $1/\alpha$ and noting that $\sup_F |\phi| = \sup_{E_2 \cap U_{j_0}} |\phi| \leq \sup_{E_2} |\phi|$, translates into

$$\|\phi\|^{\alpha}_{L^2(M)} \leq C e^{D\Lambda} \sup_{E_2} |\phi|.$$

To close the argument, we use once again the Sobolev embedding,

$$\begin{aligned} \|\phi\|_{L^\infty(M)} \leq C\|\phi\|_{\mathcal{H}^\sigma} &= \left(\sum_{\lambda_k \leq \Lambda} (1+\lambda_k)^{2\sigma} |u_k|^2 \right)^{1/2} \\ &\leq (1+\Lambda)^{\sigma} \|\phi\|_{L^2(M)} \\ &\leq C e^{D\Lambda} \sup_{E_2} |\phi|, \end{aligned}$$

which is the desired inequality.

2.1. **Control for heat equations on "very small sets".** Here we see the last 2 Theorems from Burq-Moyano's work [BM23].

Theorem 2.5 ([BM23], Theorem 3)**.** *There exists $\delta \in (0,1)$, depending only on $\dim M$, such that for any $E \subset M$ of positive $(d-\delta)$-Hausdorff measure, and any $J \subset (0,T)$ of positive Lebesgue measure, there exists $C > 0$ such that for any $u_0 \in L^2(M)$ the solution $u = e^{t\Delta}u_0$ to the heat equation with Dirichlet or Neumann boundary conditions satisfies*

{{Eq1.10}}
$$\|e^{T\Delta}u_0\|_{L^2(M)} \leq C \int_J \sup_{x \in E} |u(t,x)| dt. \tag{2.7}$$

As a consequence, under the additional assumption that E is closed subset of M, for all $u_0, v_0 \in L^2(M)$, there exists $\mu \in M((0,T) \times M)$ supported on $(0,T) \times E$ such that the solution to

{{Eq1.11}}
$$\begin{cases} (\partial_t - \Delta)u = \mu(t,x)\mathbf{1}_{J \times E}, & \mathbb{R}_+ \times M, \\ u = 0, \quad or \quad \partial_\nu u = 0, & \mathbb{R}_+ \times \partial M, \\ u(0,x) = u_0, & M, \end{cases} \tag{2.8}$$

satisfies $u\,|_{t \geq T} = e^{t\Delta}v_0$.

Theorem 2.6 ([BM23], Theorem 4)**.** *Take $\delta \in (0,1)$ as in Theorem 1 [BM23]. Let $m > 0$, $\tau \in (0,1)$ and $D > 0$. There exists $C > 0$ such that if $E_1 \subset M$ satisfies $|E_1| \geq m$ or $E_2 \subset M$ satisfies $\mathcal{C}_{\mathcal{H}}^{d-\delta}(E_2) \geq m$. Then for any sequence $(s_n)_{n \in \mathbb{N}}$,*

$$0 < \cdots < s_n < \cdots < s_0 < T,$$

converging not too fast to 0, i.e,

$$\exists \tau \in (0,1),\ \forall n \in \mathbb{N},\ (s_n - s_{n+1}) \geq \tau(s_{n-1} - s_n),$$

we have for any $u_0 \in L^2(M)$, the solution $u = e^{t\Delta}u_0$ to the heat equation,

{Eq1.11}
$$\begin{cases} \partial_t - \Delta u = 0, & \mathbb{R}_+ \times M, \\ u = 0, \quad or \quad \partial_\nu u = 0, & \mathbb{R}_+ \times \partial M, \\ u(0,x) = u_0, & M, \end{cases} \tag{2.9}$$

satisfies

{{Eq1.12}}
$$\|e^{T\Delta}u_0\|_{L^2(M)} \leq C \sup_{n \in \mathbb{N}} e^{-\frac{D}{s_n - s_{n+1}}} \int_{E_1} |e^{s_n \Delta}u_0|(s_n, x)| dx \tag{2.10}$$

and

{{Eq1.13}}
$$\|e^{T\Delta}u_0\|_{L^2(M)} \leq C \sup_{n\in\mathbb{N},x\in E_2} e^{-\frac{D}{s_n-s_{n+1}}}|e^{s_n\Delta}u_0|(s_n,x)|. \tag{2.11}$$

As a consequence, if E_1 is a closed subset of M, given any $(t_n)_{n\in\mathbb{N}}$ with $0<t_0<\cdots<t_n<\cdots<T$ converging not too fast to T. For all $u_0,v_0\in L^2(M)$, there exists (f_j) a sequence of functions on E_1 such that

$$\sum_j e^{\frac{D}{(t_{j+1}-t_j)}}\|f\|_{L^\infty(E_1)}<+\infty,$$

and the solution to

{{Eq1.15}}
$$\begin{cases}(\partial_t-\Delta)u=\sum_{j=1}^\infty \delta_{t=t_j}\otimes f_j(x)\mathbf{1}_{E_1},, & \mathbb{R}_+\times M,\\ u=0, \quad or \quad \partial_\nu u=0, & \mathbb{R}_+\times\partial M,\\ u(0,x)=u_0, & M,\end{cases} \tag{2.12}$$

satisfies $u\,|_{t>T}=e^{t\Delta}v_0$.

Similarly, under the additional assumption that E_2 is a closed subset of M, given any sequence $(t_n)_{n\in\mathbb{N}}$ with $0\leq t_0<\cdots<t_n<\cdots<T$ converging not too fast to T, for all $u_0,v_0\in L^2(M)$, there exists (μ_j) a sequence of Borel measure supported on E_2 such that

$$\sum_j e^{\frac{D}{(t_{j+1}-t_j)}}|\mu_j|(E_2)<+\infty,$$

and the solution to

{{Eq1.15}}
$$\begin{cases}(\partial_t-\Delta)u=\sum_{j=1}^\infty \delta_{t=t_j}\otimes \mu_j(x)\mathbf{1}_{E_2}, & \mathbb{R}_+\times M,\\ u=0, \quad or \quad \partial_\nu u=0, & \mathbb{R}_+\times\partial M,\\ u(0,x)=u_0, & M,\end{cases} \tag{2.13}$$

satisfies $u=|_{t>T}=e^{t\Delta}v_0$.

Since $J\subset(0,T)$ has positive Lebesgue measure, so does $J\cap(\varepsilon,T)$ for some $0<\varepsilon<T$, and hence we can assume $J\subset(\varepsilon,T)$. By subadditivity of the Hausdorff content,

$$\mathcal{C}^r_{\mathcal{H}}\left(\cup_{j\in\mathbb{N}}A_j\right)\leq\sum_{j\in\mathbb{N}}\mathcal{C}^r_{\mathcal{H}}(A_j),$$

with

$$A_j=E\cap\{x\in\mathcal{M}:\ \mathrm{dist}(x,\partial M)\geq 1/j\},\qquad j\in\mathbb{N}.$$

We deduce that there exists j_0 such that

$$\mathcal{C}^{d-\delta}_{\mathcal{H}}(E\cap A_{j_0})>0,$$

because otherwise we would have

$$\mathcal{C}^{d-\delta}_{\mathcal{H}}\left(E\cap\cup_{j\in\mathbb{N}}A_{j_0}\right)=\mathcal{C}^{d-\delta}_{\mathcal{H}}(E\setminus\partial M)=0,\quad \text{so } \mathcal{C}^{d-\delta}_{\mathcal{H}}(E)=0.$$

As a consequence, replacing E by $E\cap A_{j_0}$, we can assume that

{{Eq5.1}}
$$\exists\varepsilon>0:\forall x\in E,\quad \mathrm{dist}(x,\partial M)>\varepsilon. \tag{2.14}$$

For $w_0\in L^2(M)$ and $w=e^{(T-t)\Delta}w_0$ the solution to the backward heat equation

{{Eq5.2}}
$$\begin{cases}(\partial_t+\Delta)w=0, & (0,T)\times M,\\ w=0,\ \text{or}\ \ \partial_\nu w=0, & (0,T)\times\partial M,\\ u(T,x)=w_0, & M.\end{cases} \tag{2.15}$$

Let $\sigma>0$ such that the Sobolev embedding holds. For any $\varepsilon>0$, we have $w\in C^0([\varepsilon,T];\mathcal{H}^\sigma)$ and

$$w \in C^0([\varepsilon, T] \times M), \qquad \sup_{(t,x)\in(\varepsilon,T)\times E} |w(t,x)| \leq C\|w_0\|_{L^(M)}.$$

Consider the set

$$X = \left\{ e^{(T-t)\Delta} w_0 \mid_{J\times E} :\ w_0 \in L^2(M) \right\}.$$

We endow X with the norm inherited from $L^1((0,T); L^\infty(M))$ and have

$$\|w\|_X = \|w\|_{L^1(J;L^\infty(E))} \leq C \sup_{(t,x)\in(\varepsilon,T)\times E} |w(t,x)| \leq C\|v_0\|_{L^2(M)}.$$

By the observation estimate applied to $\tilde{J} \times E$, $\tilde{J} = T - J$, we have

{{Eq5.3}}
$$\|w \mid_{t=0} \|_{L^2(M)} \leq C \int_{\tilde{J}} \|e^{s\Delta} w_0\|_{L^\infty(E)} ds = C \int_J \|e^{(T-t)\Delta} w_0\|_{L^\infty(E)} dt \leq C\|w\|_X. \tag{2.16}$$

As a consequence, for any $u_0, v_0 \in L^2(M)$, the map

$$X \ni w \mapsto (w \mid_{t=0}, u_0 - v_0)_{L^2(M)},$$

is well defined because $w_1 = w_2 \in X$, then from (2.16), $w_1 \mid_{t=0} = w_2 \mid_{t=0}$. Also, the map from (2.16) is a continuous linear form on X and by Hahn-Banach Theorem, there exists an extension as a continuous linear form to the whole space $L^1((0,T); C^0(E))$. By the Riesz Representation Theorem, there exists $\mu \in L^\infty((0,T); \mathcal{M}(E))$ such that this linear form is given by

$$L^1((0,T); C^0(E)) \ni w \mapsto \int_{(0,T)\times E} w(t,x) d\mu.$$

We can extend μ by restriction to $L^1((0,T); C^0(M))$ in the following way:

$$L^1((0,T); C^0(M)) \ni w \mapsto \int_{(0,T)\times E} w \mid_{((0,T)\times E)} (t,x) d\mu,$$

which defines an element of $L^\infty((0,T); \mathcal{M}(M))$ denoted by μ supported on $[\varepsilon, T] \times E$, where E is closed.

Let us now check that the solution to

{{Eq5.4}}
$$\begin{cases} (\partial_t - \Delta) z = \mu(t,x) \mathbf{1}_{E\times(0,T)}, & (0,T) \times M, \\ z = 0 \text{ or } \partial_\nu z = 0, & (0,T) \times \partial M, \\ z(0,x) = 0, & M. \end{cases} \tag{2.17}$$

satisfies

$$z \mid_{t=T} = e^{T\Delta}(u_0 - v_0), \quad z \mid_{t=0} = 0,$$

and consequently choosing

$$u = e^{t\Delta} u_0 - z,$$

which proves the second part in Theorem 3. First we want to make sense of (2.17) and show that the right hand side $\mu(t,x)\mathbf{1}_{E\times(0,T)}$ is an admissible source term. We start by proving that $\mathcal{H}^\sigma$ is dense in $C^0(\mathcal{M})$ subject to Dirichlet boundary conditions, or Neumann in a sense to be specified.

{Lem5.1}
Lemma 2.2. *For every $\sigma > 0$, the set $\mathcal{H}^\sigma_D$ is dense in the set of continuous functions on $\overline{M}$ that vanishes at ∂M. $\mathcal{H}^\sigma_M$ is dense in the set of continuous functions on $\overline{M}$.*

The density of $\mathcal{H}^\sigma_N$ in C^0 implies that the map

$$\mathcal{M}(M) \ni \nu \mapsto \tilde{\nu} = \nu \mid_{\mathcal{H}^\sigma_N} \in \mathcal{H}^{-\sigma}_N,$$

is onto and consequently any measure $\nu \in \mathcal{M}(M)$ can be seen as an element of $\mathcal{H}^{-\sigma}_N$, the dual space of $\mathcal{H}^\sigma_N$. Since $\mathcal{H}^\sigma_D$ is dense in the functions that vanish on ∂M, any measure $\nu \in \mathcal{M}(M)$ supported away from the

boundary can be seen as an element of $\mathcal{H}_D^{-\sigma}$. As a consequence, we can solve (2.17) by using the natural spectral decomposition on $\mathcal{H}^{-\sigma}$,

$$\mu \sum_k \langle \mu, e_k \rangle(t) e_k,$$

with $\langle \mu, e_k \rangle(t)$ supported un (ε, T) and

$$\text{esssup}_{t\in(0,T)} \sum_k \lambda_k^{-2\sigma} |\langle \mu, e_k \rangle|^2(t) < +\infty.$$

Let $w_0 \in L^2(M)$ and let w_N be a solution to (2.15) with v_0 replaced by

$$w_{0,N} = \sum_{k\leq N} (w_0, e_k) e_k,$$

and z_N the solution to (2.17) with μ replaced by

$$\mu_N = \sum_{k\leq N} \langle \mu, e_k \rangle(t) e_k.$$

We have

$$0 = \int_0^T ((\partial_t + \Delta) w_N, z_N)_{L^2} = [(w_N, z_N)_{L^2}]_0^T - \int_0^T (w_N, (-\partial_t + \Delta) z_N)_{L^2} \tag{2.18}$$ {{Eq5.5}}

$$= (w_{0,N}, z_N \mid_{t=T})_{L^2} - \int_0^T (w_N, \mu_N)_{L^2}. \tag{2.19}$$ {Eq5.5}

We let N tend to infinity, then

$$\begin{aligned}
&w_{0,N} \to w_0, \qquad \text{in } L^2,\\
&z_N \mid_{t=T} \to z \mid_{t=T}, \quad \text{in } \mathcal{H}^{-\sigma} \text{ so } z_N \mid_{t=T} \to z \mid_{t=T} \text{ in } L^2,\\
&w_N \to w \quad \text{in } C^0([0,T]; \mathcal{H}^{\sigma}),\\
&\mu_N \to \mu, \quad \text{in } L^\infty([0,T]; \mathcal{H}^{-\sigma}).
\end{aligned}$$

We deduce that we can pass to the limit in (2.18),

$$0 = (w_0, z \mid_{t=T})_{L^2} - \int_0^T w(t,x) \mathbf{1}_{t\in(0,T)} d\mu.$$

From the defintiion of μ we have

$$\int_0^T w(t,x) d\mu = (w \mid_{t=0}, u_0 - v_0)_{L^2} = (e^{T\Delta} w_0, u_0 - v_0)_{L^2}.$$

We finally get that for all $w_0 \in L^2$,

$$(w_0, z \mid_{t=T})_{L^2} = (e^{T\Delta} w_0, u_0 - v_0)_{L^2},$$

so $z \mid_{t=T} = e^{T\Delta}(u_0, v_0)$. Thus $u = e^{t\Delta u_0} - z$ satisfies the second part of Theorem 3.

We know turn to the second part in Theorem 4 and highlight the modifications required in the previous proof. We shall focus on the case $E = E_2$ and assume that E satisfies (2.14). Let $J = \{t_n : \ n \in \mathbb{N}\} \cup \{T\} \subset [t_0, T]$ where $t_0 > 0$ and $\tilde{J} = T - J - \{s_n\} \cup \{0\}$. Let

$$X = \{e^{(T-t)\Delta} v_0 \mid_{J\times E} : \ v_0 \in L^2(M)\} \subset C^0(J \times E),$$

endowed with the sup norm. Then according to (2.11) with $s_n = T - t_n$, the linear form

$$X \ni w \mapsto (w \mid_{t=0}, u_0 - v_0)_{L^2(M)},$$

is well defined and continuous, and more precisely bounded by

$$C \sup_{n\in\mathbb{N},x\in E} e^{-\frac{B}{T-t_n}}|w(t_n,x)|. \tag{2.20}$$ {{Eq5.7}}

Indeed, since $T-t_n \geq t_{n+1}-t_n$,

$$\|w\mid_{t=0}\|_{L^2(M)} = \|e^{T\Delta}v_0\|_{L^2(M)} \leq C \sup_{n\in\mathbb{N},x\in E} e^{-\frac{B}{T-t_n}}|w(t_n,x)|.$$

By Hahn-Banach Theorem, we can extend this map to the whole space $C^0(J\times E)$, so it is still bounded by (2.20). By the Riesz Representation Theorem, this continuous linear form can be represented by a measure $\mu\in\mathcal{M}(J\times E)$ which still satisfies the same bound (2.20). As previously, we can extend this measure to a measure on $[0,T]\times M$ which is supported in $J\times E$. Hence this measure takes the form

$$\mu = \sum_n \delta_{t=t_n}\otimes\mu_n + \delta_{t=T}\times\mu_\infty,$$

with μ_j,μ_∞ measures on M supported by E. Using (2.20) we get

$$\sum_n e^{\frac{B}{T-t_n}}|\mu_n|(E) < +\infty, \quad \mu_\infty = 0.$$

Now we can make sense of solving

$$(\partial_t-\Delta)z = \sum_n \delta_{t=t_n}\otimes\mu_n, \qquad z\mid_{ty=0}=0,$$

with Dirichlet or Neumann boundary conditions in $L^\infty([0,T);\mathcal{H}^{-\sigma})$ by simply noticing that the solution to this equation is the solution to the homogeneous heat equation on (t_n,t_{n+1}) which satisfies the jump condition

$$z\mid_{t_n+0} - z\mid_{t_n-0} = \mu_n\in\mathcal{H}^{-\sigma}.$$

Since

$$\sum_n \|\mu_n\|_{\mathcal{H}^{-\sigma}} \leq C\sum_n |\mu_n|(E) < +\infty,$$

we deduce that actually $\lim_{t\to T^-} z(t)$ exists in $\mathcal{H}^{-\sigma}$ and consequently the solution exists and is unique in $[0,+\infty)$ (defined on $[T,+\infty)$ as the solution of the homogeneous heat equation). We now write the analog of the integration by parts formula (2.18). Let z_N, w_N and $\mu_{n,N}$ be the projections of z,v,μ_n on the space spanned by the first N eigenfunctions. On (t_n,t_{n+1}) we have

{{Eq5.8}}
$$0 = \int_{t_n}^{t_{n+1}} ((\partial_t+\Delta)w_N, z_N)_{L^2} = [(w_N,z_N)_{L^2}]\mid_{t_n}^{t_{n+1}} - \int_{t_n}^{t_{n+1}} (w_N,(-\partial_t+\Delta)z_N)_{L^2} \tag{2.21}$$

{Eq5.8}
$$= (w_N\mid_{t_{n+1}}, z_N\mid_{t=t_{n+1}-0})_{L^2} - (w_N\mid_{t_n}, z_N\mid_{t=t_n+0})_{L^2}, \tag{2.22}$$

which implies, using that $z_N\mid_{t=0}=0$ and $\lim_{n\to+\infty} w_N\mid_{t=t_n} = w_N(T)$ that

$$\begin{aligned}
0 &= \int_0^T ((\partial_t+\Delta)w_N, z_N)_{L^2} \\
&= \sum_n (w_N\mid_{t_{n+1}}, z_N\mid_{t=t_{n+1}-0})_{L^2} - (w_N\mid_{t_n}, z_N\mid_{t=t_n+0})_{L^2} \\
&= \lim_{k\to\infty}(w_N(t_k), z_N\mid_{t=t_k-0})_{L^2} + \sum_{n=1}^{k-1}(w_N\mid_{t_n}, z_N\mid_{t=t_n-0} - z_N\mid_{t=t_n+0})_{L^2} \\
&= (w_N(T), z_N\mid_{t=T})_{L^2} - \sum_n (w_N\mid_{t_n}, \mu_{n,N})_{L^2} \\
&= (w_N, z_N\mid_{t=T})_{L^2} - \int_0^T w_N(t)d\mu_N.
\end{aligned}$$

We can now let $N\to+\infty$ and get

$$(u_0 - v_0, z \mid_{t=T})_{L^2} = \int_0^T w(t) d\mu,$$

and we conclude as previously that $u = e^{t\Delta} u_0 - z$ satisfies (with Dirichlet or Neumann boundary conditions),

$$(\partial_t - \Delta) u = -\sum_n \delta_{t=t_n} \otimes \mu_n, \quad u \mid_{t=0} = u_0, \ u \mid_{t=T} = v_0.$$

This proves the second part in Theorem 4 in the case $E = E_2$. The case $E = E_1$ is proved similarly replacing (2.11) by (2.10).

3. Black-Box III: Unbounded domain case

In this section we follow Burq-Moyano's work [BM25], which is a generalization from the previous work to the case of non compact manifolds. The contribution is that spectral projectos are local objects.

3.1. **Setting.**

We consider on $\mathbb{R}^d$ the following Laplace operator

$$\Delta = \frac{1}{\kappa(x)} \sum_{i,j} \partial_{x_i} g^{ij}(x) \kappa(x) \partial_{x_j},$$

where we assume that the coefficients κ, g are Lipschitz,

$$\|\kappa\|_{W^{1,\infty}(\mathbb{R}^d)} + \|g\|_{W^{1,\infty}(\mathbb{R}^d)} \leq A, \tag{3.1}$$

and that g is uniformly elliptic,

$$\exists a > 0 : \forall x \in \mathbb{R}^d, \forall X \in \mathbb{R}^d, \quad a\|X\|^2 \leq (g(x)X, X), \qquad \text{and} \quad \kappa(x) \geq a. \tag{3.2}$$

Let $\omega \subset \mathbb{R}^d$ satisfying the following condition

{{Eq4}}
$$\exists R, \delta > 0, \quad \forall x \in \mathbb{R}^d, \ \text{meas}(\omega \cap B(x, R)) \geq \delta, \tag{3.3}$$

or

{{Eq5}}
$$\exists R, \delta > 0, \quad \forall x \in \mathbb{R}^d, \ \mathcal{C}^n_{\mathcal{H}}(\omega \cap B(x, R)) \geq \delta. \tag{3.4}$$

The first purpose is tho show that Jerison-Lebeau's spectral estimate from [JL99] still holds under these very weak assumptions, assuming in (3.4) that $n > d$ is sufficiently close to d, i.e, ω is not too thin.

Let us recall that $-\Delta$ on $L^2(\mathbb{R}^d, \kappa(x)dx)$ with domain $H^2(\mathbb{R}^d)$ is self-adjoint with non-negative continuous spectrum. As a consequence of the spectral theorem, is possible to write

$$u = \int_0^{+\infty} dm_\lambda u, \qquad \Pi_\mu(u) = \int_0^{\mu} dm_\lambda u, \qquad \forall u \in L^2(\mathbb{R}^d, \kappa(x)dx),$$

where $\mu > 0$ and dm_λ denotes the spectral measure of the operator $\sqrt{-\Delta}$.

Moreover, if ϕ, φ are bounded functions on $\mathbb{R}$ with real values, we have

$$\phi(\sqrt{-\Delta})u = \int_0^{+\infty} \phi(\lambda) dm_\lambda u, \quad (\phi(\sqrt{-\Delta})u, \varphi(\sqrt{-\Delta})u)_{L^2(\mathbb{R}^d, \kappa(x)dx)} = \int_0^{+\infty} \phi(\lambda)\varphi(\lambda)(dm_\lambda u, u).$$

and consequently

$$\|\phi(\sqrt{-\Delta})\Pi_\mu u\|_{L^2(\kappa dx)} \leq \sup_{\lambda \in [0,\mu]} |\phi(\lambda)| \|u\|_{L^2(\kappa dx)}. \tag{3.5}$$

Choosing $\phi_\mu - \mathbf{1}_{[0,\mu]}$, for given $\mu > 0$, we get

$$\Pi_\mu(u) = \phi_\mu(\sqrt{-\Delta}) = \mathbf{1}_{\sqrt{-\Delta} \leq \mu} u = \int_{\lambda=0}^{\mu} dm_\lambda u.$$

3.2. **Principal results.**

Theorem 3.1 ([BM25], Theorem 1)**.** *Assume that ω satisfies assumption (3.3). Then there exists $C > 0$ such that for any $u \in L^2(\mathbb{R}^d)$,*

$$\|\Pi_\mu u\|_{L^2(\mathbb{R}^d; \kappa dx)} \leq C e^{C\mu} \|\Pi_\mu u\|_{L^2(\omega; \kappa dx)}. \tag{3.6}$$

Assume now that ω verifies assumption (3.4) for some $n < d$ sufficiently close to d. Then there exists $C > 0$ such that for any $u \in L^2(\mathbb{R}^d)$,

$$\|\Pi_\mu u\|^2_{L^2(\mathbb{R}^d; \kappa dx)} \leq C e^{C\mu} \sum_{k \in \mathbb{Z}} \|\Pi_\mu u\|^2_{L^\infty(\omega \cap B(k,R))}. \tag{3.7}$$

Now we give three applications to the observation and control of the heat equation.

Theorem 3.2 ([BM25],Theorem 2.). *Let $F \subset (0,T)$ be a set of positive measure. Let $E \subset \mathbb{R}^d$ satisfying (3.3). Then there exists $C>0$ such that for any $u_0 \in L^2(\mathbb{R}^d)$, the solution $u = e^{t\Delta}u_0$ to the heat equation*

$$\partial_t u - \Delta u = 0, \qquad u\,|_{t=0} = u_0,$$

satisfies

$$\|e^{T\Delta}u_0\|^2_{L^2(\mathbb{R}^d)} \leq C \int_{F\times E} |u(t,x)|^2 \kappa(x) dxdt. \tag{3.8}$$

As a consequence, for all $u_0, v_0 \in L^2(\mathbb{R}^d)$ there exists $f \in L^2(F\times E)$ such that the solution to

$$(\partial_t - \Delta)u = f\mathbf{1}_{F\times E}(t,x), \quad u\,|_{t=0} = u_0,$$

satisfies

$$u\,|_{t=T} = e^{T\Delta}v_0.$$

The second application is a control and observability result from sets of positive Hausdorff content.

Theorem 3.3 ([BM25], Theorem 3.). *Let $F \subset (0,T)$ be a set of positive measure. Let $E \subset \mathbb{R}^d$ satisfying (3.4). Then, there exists $C>0$ such that for any $u_0 \in L^2(M)$, the solution $u = e^{t\Delta}u_0$ to the heat equation*

$$\partial_t u - \Delta u = 0, \quad u\,|_{t=0} = u_0,$$

satisfies

$$\|e^{T\Delta}u_0\|^2_{L^2(\mathbb{R}^d)} \leq C \sum_{k\in\mathbb{Z}^d} \int_F \sup_{x\in E\cap B(k,R)} |u(t,x)|^2 dt. \tag{3.9}$$ {{Eq10}}

Remark 3.1. *Denote by G the set of function on F with values in the set of locally bounded Radon measures on $\overline{E}$, such that*

$$\|f\|^2_G = \sum_{k\in\mathbb{Z}^d} \int_F |f(\overline{E}\cap B(k,R))|^2(t)dt < +\infty.$$

Notice that G is the dual space of $\tilde{G}$, defined as the space of functions g on F with values continuous functions on $\overline{E}$ such that

$$\|g\|^2_{\tilde{G}} = \sum_{k\in\mathbb{Z}^d} \int_F \sup_k |f(E\cap B(k,R))|^2 dt < +\infty.$$

We deduce from (3.9) that for all $u_0, v_0 \in L^2(\mathbb{R}^d)$, there exists $f \in G$ such that the solution to

$$(\partial_t - \Delta)u = f\mathbf{1}_{F\times\overline{E}}(t,x), \quad u\,|_{t=0} = u_0, \tag{3.10}$$ {{Eq11}}

satisfies $u\,|_{t=T} = e^{T\Delta}v_0$.

Remark 3.2. *For $\chi \in C^\infty_0(\mathbb{R}^d)$, from the Sobolev injection [BM23, Proposition 2.1], for $\sigma>0$ large enough, the map*

$$u \in D(-\Delta)^\sigma \mapsto \chi u \in C^0,$$

is well defined. Hence by duality, the map

$$v \in \mathcal{M} \mapsto \chi v \in D(-\Delta)^{-\sigma},$$

is also well defined and continuous. As a consequence, G is continuously embedded in

$$\mathcal{H} = L^2(F; D(-\Delta)^{-\sigma}),$$

and equation (3.10) is just solved by

$$u = e^{t\Delta}u_0 + \int_0^t e^{(t-s)} f\mathbf{1}_{F\times\overline{E}}(s,x)ds.$$

The third application is the observation and control for discrete times from [BM25, Theorem 4].

Theorem 3.4 ([BM25][Theorem 4)**.** , *Observability and exact controllability using controls localised at fixed times] Let $m > 0, \tau \in (0,1)$ and $D > 0$. There exists $C > 0$, such that if $E \subset \mathbb{R}^d$ satisfies (3.3), then for any sequence $(s_n)_{n\in\mathbb{N}}$,*

$$J = \{0 < \cdots < s_n < \cdots < s_0 < T\},$$

converging not too fast to 0, i.e,

$$\exists \tau \in (0,1), \forall n \in \mathbb{N}, (s_n - s_{n+1}) \geq \tau(s_{n-1} - s_n),$$

we have that for any $u_0 \in L^2(M)$, the solution $u = e^{t\Delta}u_0$ to the eat equation

$$\partial_t u - \Delta u = 0, \quad u\mid_{t=0} = u_0,$$

satisfies

$$\|e^{T\Delta}u_0\|^2_{L^2(M)} \leq C \sup_{n\in\mathbb{N}} e^{-\frac{D}{s_n - s_{n+1}}} \int_E |e^{s_n\Delta}u_0|^2(s_n, x)dx. \tag{3.11}$$

As a consequence, given any sequence $(t_n)_{n\in\mathbb{N}}$,

$$J = \{0 < t_0 < \cdots < t_n < \cdots < T\},$$

converging not too fast to T,

$$\exists \tau \in (0,1), \forall n \in \mathbb{N}, (t_{n+1} - t_n) \geq \tau(t_n - t_{n-1}), \tag{3.12}$$

for all $u_0, v_0 \in L^2(M)$ there exists $(f_j)_j$ a sequence of functions on E_1 such that

$$\sum_j e^{\frac{D}{(t_{j+1}-t_j)}} \|f\|_{L^2(E)} < +\infty,$$

and the solution to

$$(\partial_t - \Delta)u = \sum_{j=1}^{\infty} \delta_{t=t_j} \otimes f_j(x)\mathbf{1}_{E_1}, \tag{3.13}$$

satisfies $u\mid_{t=T} = e^{t\Delta}v_0$.

Remark 3.3. *In [BM23], the spectral estimates are obtained in manifolds with boundary under Dirichlete or Neumann boundary conditions. It's possible to include such boundaries in a non compact setting provided we still control at infinity.*

3.3. **Proof of Theorem 1.** Let us define

$$v_\mu(t,x) = \int_{\lambda=0}^{\mu} \frac{\sinh(\lambda t)}{\lambda} dm_\lambda u. \tag{3.14}$$

Recall that

$$-\Delta v_\mu = \int_{\lambda=0}^{\mu} \frac{\sinh(\lambda t)}{\lambda} \lambda^2 dm_\lambda u.$$

As a consequence, we have

$$\left(\frac{1}{\kappa} \operatorname{div} g^{-1}\kappa\nabla_x + \partial_t^2\right) v_\mu = 0 \Longrightarrow (\operatorname{div} g^{-1}\kappa\nabla_x + \partial_t\kappa\partial_t)v_\mu = 0.$$

Without loss of generality, we can asume that in (3.3) and (3.4), the constant R is chosen large enough so that

$$\cup_{p\in\mathbb{Z}^d} B(p,R) = \mathbb{R}^d.$$

Let $1 = \sum_p \chi(x-p)$ be a partition of unity associated. Consider, for $T_2 > T_1 > 0$ the sets

$$K_p := [-T_1, T_1] \times \overline{B(p,R)}, \qquad \Omega_p := (-T_2, T_2) \times B(p, 2R),$$
$$F_p := \omega \cap B(p,R), \qquad E_p := \{0\} \times F_p,$$

which by construction satisfy the inclusions $E_p \subset K_p \subset \Omega_p$. Now assumption (3.3) ensures that $|F_p| \geq \delta$. Note that by Hausdorff measure, since F_p has a diameter at most $2R$, when computing the Hausdorff content of F_p, we can assume that for every j, we have $r_j \leq 4R$. As a consequence,

$$\mathcal{C}^{d-1-\varepsilon}(E_p) \geq (4R)^{-\varepsilon}\mathcal{C}^{d-1}(E_p) \geq c\delta.$$

For sufficiently small $\varepsilon > 0$, we can now apply [LM18, Theorem 5.1] and get

$$\sup_{K_p} |\nabla_{t,x} v_\mu| \leq C \left(\sup_{E_p} |\nabla_{t,x} v_\mu| \right)^{\alpha} \left(\sup_{\Omega_p} |\nabla_{t,x} v_\mu| \right)^{1-\alpha} \tag{3.15}$$ {{Eq16}}

where the constant is uniform with respecto to p because the metric and κ are bounded on $B(p, 2R)$. (In fact, uniformly with respect to p while the metric is uniformly elliptic).

First we replace the L^∞ norm by L^2 norm. we can assume $v_\mu \neq 0$, let

$$a = \left(\varepsilon \frac{\sup_{K_p} |\nabla_{t,x} v_\mu}{\sup_{\Omega_p} |\nabla_{t,x} v_\mu|} \right)^{1/a}, \quad E_p' = \{x \in E_p : |\nabla_{t,x} v_\mu| \leq a\}.$$

Then for $\varepsilon C < 1, |E_p'| \leq \delta/2$. Indeed, otherwise (3.15) would hold with E_p replaced by E_p' and the same constants, leading to

$$\sup_{K_p} |\nabla_{t,x} v_\mu| \leq C(a)^\alpha \left(\sup_{\Omega_p} |\nabla_{t,x} v_\mu| \right)^{1-\alpha} \implies \sup_{K_p} |\nabla_{t,x} v_\mu| \leq C\varepsilon \sup_{K_p} |\nabla_{t,x} v_\mu|, \tag{3.16}$$ {{Eq17}}

which, if $C\varepsilon < 1$, implies $\nabla_{t,x} v_\mu \,|_{K_p} \equiv 0$, hence $\nabla_{t,x} v_\mu$ is identically 0.

As a consequence,

$$\int_{E_p} |\nabla_{t,x} v_\mu|^2 dx \geq \int_{E \setminus E_p'} |\nabla_{t,x} v_\mu|^2 dx \geq a^2 \frac{\delta}{2} \geq \frac{\delta}{2} \left(\varepsilon \frac{\sup_{K_p} |\nabla_{t,x} v_\mu|}{(\sup_{\Omega_p} |\nabla_{t,x} v_\mu|)^{1-\alpha}} \right)^{2/\alpha},$$

and we deduce

$$\|\nabla_{t,x} v_\mu\|_{L^2(B(p,R))} \leq C \left(|\sup_{\Omega_p} |\nabla_{t,x} v_\mu| \right)^{1-\alpha} \|\nabla_{t,x} v_\mu\|^{\alpha}_{L^2(E \cap B(p,R))}. \tag{3.17}$$ {{Eq18}}

To end the proof of the first part, we need the following variant of Sobolev embedding.

Lemma 3.1 ([BM23], Lemma 2.1)**.** *There exists $C, K > 0$ such that for all $p \in \mathbb{Z}^d$, and all $u \in L^2(\mathbb{R}^d; \kappa(x)dx)$,*

$$\left(\sum_{p \in \mathbb{Z}^d} \|\nabla_{t,x} v_\mu\|^2_{L^\infty((-T_2,T_2) \times B(p,2R))} \right)^{1/2} \leq C e^{K\mu} \|\Pi_\mu u\|_{L^2(\mathbb{R}^d; \kappa(x)dx)}.$$

Assuming the previous lemma, observe that Young's inequality yields

$$a^{2-2\alpha} b^{2\alpha} \leq C(a^2 + b^2),$$

hence, from (3.17) we deduce that for all $D > 0$,

$$\|\Pi_\mu u\|^2_{L^2(B(p,R))} \leq C e^{-D\mu} \sup_{\Omega_p} |\nabla_{t,x} v_\mu|^2 + C e^{\frac{2-2\alpha}{2\alpha} D\mu} \|\Pi_\mu u\|^2_{L^2(E \cap B(p,R))}.$$

Now, from the previous lemma we deduce

$$\begin{aligned}\|\Pi_\mu u\|^2_{L^2(B(p,R))} &\leq Ce^{-D\mu}\sum_{p\in\mathbb{Z}}\sup_{\Omega_p}|\nabla_{t,x}v_\mu|^2 + Ce^{D\frac{2-2\alpha}{2\alpha}\mu}\sum_{p\in\mathbb{Z}}\|\Pi_\mu u\|^2_{L^2(E\cap B(p,R))}\\ &\leq Ce^{-D\mu}e^{2K\mu}\|\Pi_\mu u\|^2_{L^2(\mathbb{R}^d)} + Ce^{D\frac{2-2\alpha}{2\alpha}\lambda}\|\Pi_\mu u\|^2_{L^2(E)},\end{aligned}$$

and the proof of Theorem 1 for μ large enough follows from taking $D > 2K$.

Now, for the second case we have from assumption (3.4),

$$\mathcal{C}^n(E_p) \geq \delta,$$

and we deduce from [LM18, Theorem 5.1], if n is sufficiently close to d, that

$$\sup_{K_p}|\nabla_{t,x}v_\mu| \leq C\left(\sup_{E_p}|\nabla_{t,x}v_\mu|\right)^\alpha\left(\sup_{\Omega_p}|\nabla_{t,x}v_\mu|\right)^{1-\alpha},$$

where the constant is uniform with respect to p because the metric and κ are bounded on $B(p,2R)$, since the metric is uniformly elliptic. We deduce from $\Pi_\mu u = \partial_t v_\mu\,|_{t=0}$,

$$\|\Pi_\mu u\|_{L^2(B(p,R))} \leq C\sup_{K_p}|\nabla_{t,x}v_\mu| \leq C\left(\sup_{E_p}|\nabla_{t,x}v_\mu|\right)^\alpha\left(\sup_{\Omega_p}|\nabla_{t,x}v_\mu|\right)^{1-\alpha},$$

and we deduce for all $D > 0$,

$$\|\Pi_\mu u\|^2_{L^2(B(p,R))} \leq Ce^{-D\mu}\sup_{\Omega_p}|\nabla_{t,x}v|^2 + Ce^{\frac{2-2\alpha}{2\alpha}D\mu}\|\Pi_\mu u\|^2_{L^\infty(E\cap B(p,R))}.$$

Now, from the Sobolev embedding we deduce

$$\begin{aligned}\|\Pi_\mu u\|^2_{L^2(\mathbb{R}^d)} &\leq Ce^{-D\mu}\sum_{p\in\mathbb{Z}}\sup_{\Omega_p}|\nabla_{t,x}v|^2 + Ce^{D\frac{2-2\alpha}{2\alpha}\mu}\sum_{p\in\mathbb{Z}}\|\Pi_\mu u\|^2_{L^\infty((E\cap B(p,R))\mathbb{R}^d)}\\ &\leq Ce^{-D\mu}e^{2K\mu}\|\Pi_\mu u\|^2_{L^2(\mathbb{R}^d)} + Ce^{D\frac{2-2\alpha}{2\alpha}\lambda}\sum_{p\in\mathbb{Z}^d}\|\Pi_\mu u\|^2_{L^\infty(E\cap B(p,R))},\end{aligned}$$

and the proof of the second part for $\mu \geq \mu_0$ follows from taking $D > 2K$. Finally, for $\mu < \mu_0$, apply Theorem 1 for μ_0 to $u = \Pi_\mu(u)$.

4. 6/8: Double manifold and further results

Here we collect some comments about the double manifold an its consequences in some control/observability problems. The construction is interesting as a concept, so we recall some good properties before going further into some applications.

4.1. **Black-Box IV: The double manifold.** Suppose that we have a manifold with boundary M and Dirichlet or Neumann boundary conditions on ∂M. The classical idea is to reduce this questions to the case of a manifold without boundary by gluing two copies of M along the boundary in such a way that the new double manifold $\tilde{M}$ inherits a Lipschitz metric, which allows one to apply the previous results for boundaryless manifolds for this double manifold.

The gluing process has to be done properly, as otherwise the resulting glued metric might not even be continuous. The main difficulty in our context comes from the fact that the usual method for this doubling procedure relies on the use of a reflection principle in geodesic coordinate systems. The existence of such coordinate system requires at least C^2 (resp. C^3) regularity for the metric (resp. the domain), compared to our $W^{1,\infty}$ and $W^{2,\infty}$ assumptions. In other words, with this low regularity level we do not expect to get a C^1, and hence integrable, geoedesic flow.

To circumvent this technical difficulty, we shall define a pseudo-geodesic system relying on a regularization of the normal direction of the boundary, which will be $W^{2,\infty}$ and tangent at the boundary to the "geodesic coordinate system", which actually does not exist at this low regularity level.

Let $\tilde{M} = \overline{M} \times \{-1, 1\}/\partial M$ be the double space made up of two copies of $\overline{M}$ where we identify the points on the boundary, $(x, -1)$ and $(x, 1)$ for $x \in \partial M$.

Theorem 4.1. *The double manifold, [BM23] Let g be given. There exists a $W^{2,\infty}$ structure on the double manifold $\tilde{M}$, a metric $\tilde{g}$ of class $W^{1,\infty}$ on $\tilde{M}$ and a density $\tilde{\kappa}$ of class $W^{1,\infty}$ on $\tilde{M}$ such that the following hold.* {Thm7}

1. *The maps*
$$\iota^{\pm} : M \ni x \mapsto (x, \pm 1) \in \tilde{M} = M \times \{1, -1\}/\partial M,$$
are isometric embeddings.
2. *The density induced on each copy of M is the density κ.*
3. *For any eigenfunction e with eigenvalue λ^2 of the Laplace $-\Delta = -\kappa^{-1} \operatorname{div} G^{-1}\kappa\nabla$ operator with Dirichlet or Neumann boundary conditions, there exists an eigenfunction $\tilde{e}$ with the same eigenvalue λ of the Laplace operator $-\Delta = -(\tilde{\kappa})^{-1} \operatorname{div} \tilde{g}\tilde{g}^{-1}\tilde{\kappa}\nabla$ on $\tilde{M}$ such that*
$$\tilde{e}\,|_{M\times\{1\}} = e, \quad \tilde{e}\,|_{M\times\{-1\}} = \begin{cases} -e, & \text{Dirichlet BC,} \\ e, & \text{Neumann BC,} \end{cases} \tag{4.1}$$ {Eq3.1}

Corollary 4.1. *The estimate for bondaryless manidols,*
$$\phi = \sum_{\lambda_k \leq \Lambda} u_k e_k(x) \Longrightarrow \|\phi\|_{L^\infty(M)} \leq C e^{D\Lambda} \sup_{x \in E_2} |\phi(x)|,$$
implies
$$\phi = \sum_{\lambda_k \leq \Lambda} u_k e_k(x) \Longrightarrow \|\phi\|_{L^\infty(M)} \leq C e^{D\Lambda} \|\phi \mathbf{1}_{E_1}\|_{L^1(M)}.$$
for Dirichlet or Neumann boundary conditions, and in the Dirichlet case we can even add any constant to the spectral projector and replace ϕ by
$$\Psi = u_0 + \sum_{\lambda_k \leq \Lambda} u_k e_k(x).$$

Remark 4.1. *Since orthonormal basis for the Dirichlet/Neumann Laplacian are dense in $L^2(M)$, the vector space generated by their extensions as defined in (4.1) is dense in $L^2(\tilde{M})$, concluding the existence of a Hilbert basis of $L^2(\tilde{M})$ made up of eigenfunctions of $\tilde{\Delta}$ on $\tilde{M}$.*

Proposition 4.1. *[BM23], Proposition 3.4 Assume that M is a $W^{2,\infty}$ manifold of dimension d with boundary endower with a Lipschitz positive definite metric g and a Lipschitz positive density κ. Let*

$$\Delta = \frac{1}{\kappa} \operatorname{div} g^{-1}(x)\kappa\nabla_x = \operatorname{div}_\kappa \nabla_g.$$

Then near any point $X_0 \in \partial M$, there exists a $W^{2,\infty}$ coordinate system

$$X_0 = (0,0) \in \mathbb{R}_y \times \mathbb{R}_z^{d-1}, \tag{4.2}$$

$$\Omega = (0,+\infty) \times \mathbb{R}^{d-1}, \quad \partial\Omega = \{0\} \times \mathbb{R}^{d-1}, \tag{4.3}$$

$$\Delta = \frac{1}{\kappa(y,z)} \nabla_{y,z} \kappa(y,z)^\top b(y,z) \nabla_{y,z}, \quad b\mid_{\partial\Omega} = \begin{pmatrix} 1 & 0 \\ 0 & b'(z) \end{pmatrix}. \tag{4.4}$$

Here, $b(s,z)$ denotes the normal metric given for $s \geq 0$ by

$$b(s,z) = (\psi_j' a)^\top a(y,z) \psi_j'.$$

Remark 4.2. *In a geodesic coordinate system, we would have ad iagonal form for the metric in a neighbourhood of the bounday. The previous proposition corresponds to the fact that our coordinate system is, at the boundary, "tangent to a geodesic coordinate system".*

Summarizing, we have defined a covering $\cup_{j=1}^N U_j' \supset \partial M$ and $W^{2,\infty}$ diffeomorphisms,

$$\psi_j' : U_j' \to \mathbb{R}^d, \quad \mathbb{R}_s \times \mathbb{R}_x^{d-1},$$

such that $\psi_j'(V_j) \subset B(0,1)_{s,x} \cap \{s \geq 0\}$, and after the change of variables ψ_j', the metric takes the previous form at the boundary.

Now we perform the gluing by defining a covering of ∂M, seen as a subset of $\tilde{M}$.

$$\partial M \subset \cup_{j=1}^N U_j' \times \{-1,1\} = \cup_{j=1}^N \tilde{U}_j,$$

where we identify the points in $U_j' \cap \partial M \times \{-1,1\}$, and define the map

$$\Psi_j : U_j' \times \{\varepsilon\} \ni z \mapsto \begin{cases} \psi_j'(x), & \varepsilon = 1, \\ S \circ \psi_j'(x), & \varepsilon = -1, \end{cases}$$

where $S(s,z) = (-s,z)$.

To conclude the first part of Theorem 4.1, it remains to check that

(1) The image of the metric on $\tilde{M}$ induced by the metrics on the two copies of $\overline{M}$ is well defined and Lipschitz,
(2) The changes of charts
$$\Psi_k \circ \Psi_j^{-1} : \Psi_j(\tilde{U}_k \cap \tilde{U}_j) \to \mathbb{R}^d,$$
are $W^{2,\infty}$,
(3) The density $\tilde{\kappa}$ obtained by gluing two copies of κ on each copy of M is $W^{1,\infty}$.

One could check [BM23, Section 3] to see the calculations and proofs of the previous claims. Finally, to prove the second part about the eigenfunctions, let e be an eigenfunction of $-\Delta$ on M with Dirichlet boundary conditions associated to the eigenvalue λ^2. We define

$$\tilde{e}(x, \pm 1) = \pm e(x).$$

This definition makes sense because $e(x) = 0 = -e(x)$ on the boundary. Now we check that $\tilde{e}$ is the eigenfunction of $\tilde{\Delta}$ on $\tilde{M}$. Away from the boundary ∂M this is clear, while near a point $x \in \partial M \subset \tilde{M}$ we can work in a coordinate chart $(\Psi_j, \tilde{U}_j)$. In this chart,

$$\tilde{e}(s,z) = \begin{cases} e(s,z), & s > 0, \\ -e(-s,z), & s < 0. \end{cases}$$

Followin some chart-calculations, the case is proved. Neumann case follows similarly defining

$$\tilde{e}(s,z) = \begin{cases} e(s,z), & s>0, \\ e(-s,z), & s<0. \end{cases}$$

4.2. **Simultaneous control for the Heat equation with Dirichlet and Neumann boundary conditions.** Consider a smooth bounded domain $\Omega \subset \mathbb{R}^d$ and let $\omega \subset \Omega$ be a subset of positive measure $|\omega| > 0$ and the following internal simultaneous controlability problem

$$\begin{cases} (\partial_t - \Delta)u = f\mathbf{1}_{(0,T)\times\omega}, & u\,|_{\partial\Omega} = 0, & u\,|_{t=0} = u_0, \\ (\partial_t - \Delta)v = f\mathbf{1}_{(0,T)\times\omega}, & \partial_\nu u\,|_{\partial\Omega} = 0, & v\,|_{t=0} = v_0, \end{cases} \tag{4.5}$$ {{eq: 1.1}}

Definition 4.1. *We shall say that the heat equation in Ω is simultaneously null controllable with Dirichlet and Neumann boundary conditions if for any $u_0, v_0 \in L^2(\Omega)$, there exists $f \in L^2((0,T)\times\omega)$ such that the solution of the system (4.3) satisfies*

$$u\,|_{t>T} = 0, \quad v\,|_{t>T} = 0.$$

Remark 4.3. *The interest of this problem lies on the fact that both heat equations exhibit relatively independent dynamics and yet they can be steered to zero using exactly the same control.*

{BMSimul: Thm

Theorem 4.2. *Let $T>0$, $\omega \subset \Omega$ of positive measure $|\omega|>0$. Then the heat equation in Ω is simultaneously null controlable with Dirichlet and Neumann boundary conditions.*

Remark 4.4. *This simultaneous controlability seems counter intuitive since no coupling exists between these two equations. Yet by considering $w_1 = u+v$ and $w_2 = u-v$, the problem is reduced to the controllability of (w_1, w_2) with a control acting only in w_1 and the system inherit two transmission conditions. Here we will not use this path but the double manifold technique.*

Denote by e^D_λ (resp. e^N_μ) the spectral family associated to the Laplacian in Ω with Dirichlet (resp. Neumann) conditions. Given $\omega \subset \Omega$ with $|\omega| > 0$, there exists constants $C, D > 0$ such that

$$\|\Pi^D_\Lambda u\|_{L^\infty}(\Omega) \leq Ce^{C\Lambda}\|\mathbf{1}_\omega \Pi^D_\Lambda u\|_{L^1(\Omega)}, \quad \forall u \in L^2(\Omega),$$

and

$$\|\Pi^N_\Lambda u\|_{L^\infty}(\Omega) \leq Ce^{C\Lambda}\|\mathbf{1}_\omega \Pi^N_\Lambda u\|_{L^1(\Omega)}, \quad \forall u \in L^2(\Omega).$$

The second theorem shows that these spectral inequalities also hold simultaneously,

{BMSimul: Thm

Theorem 4.3. *Let $\omega \subset \Omega$ with $|\omega| \geq m$. There exists $C, D > 0$ such that for any $\Lambda > 0$, we have*

$$\|\Pi^D_\Lambda u\|_{L^\infty(\Omega)} + \|\Pi^N_\Lambda v\|_{L^\infty(\Omega)} \leq Ce^{C\Lambda}\|\mathbf{1}_\omega(\Pi^D_\Lambda u + \Pi^N_\Lambda v)\|_{L^1(\omega)}, \qquad \forall u, v \in L^2(\Omega). \tag{4.6}$$

Before proving the theorem, let us denote by $\tilde{\Pi}_\Lambda$ the spectral projector on the manifold $\tilde{M}$. Let $u, v \in L^2(M)$ and define the function

$$\tilde{u}(x,1) = (u+v)(x), \qquad \tilde{u}(x,-1) = (-u+v)(x). \tag{4.7}$$ {{BMSimul: eq

Clearly we have

$$u = \sum_k u_k e^D_k, \quad v = \sum_k v_k e^N_k \Longrightarrow \tilde{u} = \sum_k u_k \tilde{e}^D_k + v_k \tilde{e}^N_k.$$

Thanks to the reflection principle from the double manifold, we can link the Dirichlet and Neumann spectral projectors on M and the spectral projector on $\tilde{M}$ by the relation

$$\tilde{\Pi}_\Lambda(\tilde{u})(\cdot,1) = \Pi^D_\Lambda(u(\cdot)) + \Pi^N_\Lambda(v(\cdot)), \qquad \tilde{\Pi}_\Lambda(\tilde{u})(\cdot,-1) = -\Pi^D_\Lambda(u(\cdot)) + \Pi^N_\Lambda(v(\cdot)). \tag{4.8}$$ {{BM23eq: 2.3

By the previous work,

Theorem 4.4. *[BM23], Theorem 1. Let $\tilde{\omega} \subset \tilde{M}$ with positive Lebesgue measure. Then, there exists $C > 0$ such that for any $\Lambda > 0$ and any $\tilde{u} \in L^2(\tilde{M})$ we have*

$$\|\tilde{\Pi}_\Lambda \tilde{u}\|_{L^\infty(\tilde{M})} \leq Ce^{C\Lambda} \|\mathbf{1}_{\tilde{\omega}} \tilde{\Pi}_\Lambda \tilde{u}\|_{L^1(\tilde{\omega})}.$$

Now we have all the needed ingredients.

Proof: Proof (4.3) Let $\omega \subset M$ with positive Lebesgue measure. Let $\tilde{\omega} = \omega \times \{1\}$. According to [BM23, Theorem 1] and (**??**), we get for any $u, v \in L^2(M)$,

$$\begin{aligned}
\|\Pi^D_\Lambda u\|^2_{L^\infty(M)} + \|\Pi^N_\Lambda v\|^2_{L^\infty(M)} &= \|\Pi^D_\Lambda u + \Pi^N_\Lambda v\|_{L^\infty(M)} && (4.9)\\
&= \|\tilde{\Pi}_\Lambda \tilde{u}\|^2_{L^\infty(\tilde{M})} && (4.10)\\
&\leq Ce^{C\Lambda} \|\mathbf{1}_{\tilde{\omega}} \tilde{\Pi}_\Lambda \tilde{u}\|^2_{L^1(\tilde{\omega})} && (4.11)\\
&= Ce^{C\Lambda} \|\mathbf{1}_{\omega\times\{1\}\tilde{\Pi}_\Lambda \tilde{u}}\|^2_{L^1(\tilde{\omega})} && (4.12)\\
&= Ce^{C\Lambda} \|\mathbf{1}_\omega \Pi^D_\Lambda u + \Pi^N_\Lambda v\|^2_{L^1(\omega)}. && (4.13)
\end{aligned}$$

□

4.3. **Control and the double manifold.** By the previous Burq-Moyano's work [BM23], we have the next result adapted to our context

Theorem 4.5. *Let $\tilde{\omega} \subset \tilde{M}$ be a measurable set with $|\tilde{\omega}| > 0$. Then for every $T > 0$ and every $\tilde{u}_0 \in L^2(M)$, there exists $\tilde{f} \in L^2((0,T) \times \tilde{\omega})$ such that the solution to the heat equation on $\tilde{M}$ satisfies*

$$\tilde{u} \mid_{t \geq T} = 0.$$

Now we prove Theorem (4.2).

Proof: For any $u, v \in L^2(M)$ and define $\tilde{u}$ by (4.5), and for any $\omega \subset M$ of positive measure, let $\tilde{\omega} = \omega \times \{1\}$. According to the previous theorem, for every $T > 0$, there exists $f \in L^2((0,T) \times \tilde{\omega})$ such that

3eq: 2.6}} $$(\partial_t - \tilde{\Delta})\tilde{U} = \tilde{f}\mathbf{1}_{(0,T)\times\omega}, \quad \tilde{U} \mid_{t=0} = \tilde{u}, \quad \tilde{U} \mid_{t \geq T} = 0. \tag{4.14}$$

Let us define next

$$u(x,t) = \tilde{U}(t,x,1) - \tilde{U}(t,x,-1), \qquad v(t,x) = \tilde{U}(t,x,1) + \tilde{U}(t,x,-1),$$

and notice that u (resp. v) satisfies the Dirichlet (resp. Neumann) boundary conditions. The second one is harder to verify but comes from the double manifold construction.

By the same construction, we define a normal coordinate system near any point in the boundary of M such that $M = \{x_n > 0\}$ and then we glued the two copies defined by

$$M \times \{1\} = \{x_n > 0\}, \qquad M \times \{-1\} = \{x_n < 0\},$$

with the relation

$$(x,1) = (x_n, x', 1), \qquad (x,-1) = (-x_n, x', -1),$$

which implies

$$\partial_\nu v = \partial_{x_n} \left(\tilde{U}(t,x,1) + \tilde{U}(t,x,-1)\right) \mid_{x_n=0} = \partial_{x_n}(\tilde{U})(t,x,1) - \partial_{x_n}(\tilde{U})(t,x,-1) = 0.$$

now by definition of u and v we have

$$(\partial_t - \Delta)u = \tilde{f}(t,x,1)\mathbf{1}_{(0,T)\times\omega} - \tilde{f}(t,x,-1)\mathbf{1}_{(0,T)\times\tilde{\omega}} = f(t,x)\mathbf{1}_{(0,T)\times\omega},$$

as $\tilde{f}(t,x,-1)\mathbf{1}_{(0,T)\times\tilde{\omega}} = 0$ by the choice $\tilde{\omega} = \omega \times \{1\}$. By the same idea,

$$(\partial_t - \Delta)v = \tilde{f}(t,x,1)\mathbf{1}_{(0,T)\times\omega} + \tilde{f}(t,x,-1)\mathbf{1}_{(0,T)\times\overline{\omega}} = f(t,x)\mathbf{1}_{(0,T)\times\omega}.$$

As a consequence, u and v solve (4.3) with control $f\mathbf{1}_{(0,T)\times\omega}$. Finally using (4.12) we get

$$u\,|_{t\geq T} = 0, \qquad v\,|_{t\geq T} = 0,$$

which ends the proof. □

References

[BM23] Nicolas Burq and Iván Moyano. "Propagation of smallness and control for heat equations". English. In: *J. Eur. Math. Soc. (JEMS)* 25.4 (2023), pp. 1349–1377. ISSN: 1435-9855. DOI: 10.4171/JEMS/1213.

[BM25] Nicolas Burq and Ivan Moyano. "Propagation of smallness and spectral estimates". English. In: *J. Anal. Math.* 156.1 (2025), pp. 35–46. ISSN: 0021-7670. DOI: 10.1007/s11854-025-0374-3.

[JL99] David Jerison and Gilles Lebeau. "Nodal sets of sums of eigenfunctions". English. In: *Harmonic analysis and partial differential equations. Essays in honor of Alberto P. Calderón's 75th birthday. Proceedings of a conference, University of Chicago, IL, USA, February 1996*. Chicago, IL: The University of Chicago Press, 1999, pp. 223–239. ISBN: 0-226-10456-7.

[LM18] Alexander Logunov and Eugenia Malinnikova. "Quantitative propagation of smallness for solutions of elliptic equations". English. In: *Proceedings of the international congress of mathematicians 2018, ICM 2018, Rio de Janeiro, Brazil, August 1–9, 2018. Volume III. Invited lectures*. Hackensack, NJ: World Scientific; Rio de Janeiro: Sociedade Brasileira de Matemática (SBM), 2018, pp. 2391–2411. ISBN: 978-981-327-292-7; 978-981-327-287-3; 978-981-3272-89-7. DOI: 10.1142/9789813272880_0143.

[RLR22] Jérôme Le Rousseau, Gilles Lebeau, and Luc Robbiano. "Elliptic Carleman Estimates and Applications to Stabilization and Controllability, Volume I". In: (Jan. 2022). DOI: 10.1007/978-3-030-88674-5. URL: https://doi.org/10.1007/978-3-030-88674-5.